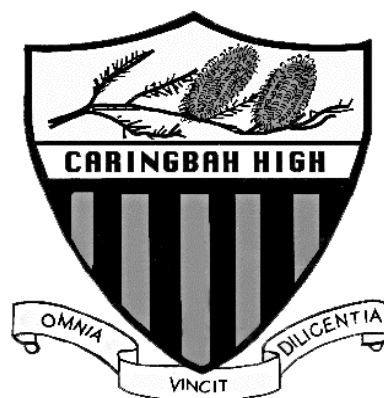




Caringbah High School

Year 12 2022

Mathematics Extension 2

HSC Course

Assessment Task 1

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 6-9, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 40

Section I 5 marks

Attempt Questions 1-5
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 35 marks

Attempt Questions 6-9
Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only						
Section I	Section II				Total	
Q 1-5	Q6	Q7	Q8	Q9		
/5	/9	/8	/9	/9	/40	%

Section I

5 marks

Attempt Questions 1-5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5

1. What is $-1 + i\sqrt{3}$ in modulus-argument form?

(A) $2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

(B) $\sqrt{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

(C) $2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$

(D) $\sqrt{2}\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$

2. Consider the statement:

P : For every integer x , there exists an integer y such that $y^2 > x$.

Which of the following statements is $\neg P$?

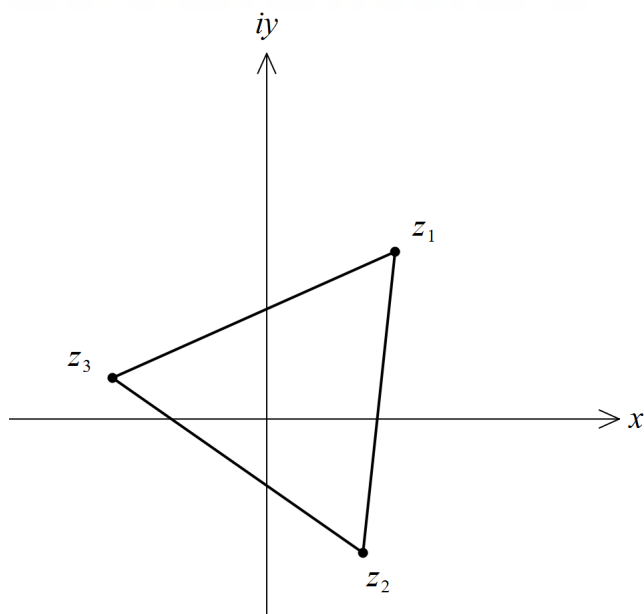
(A) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, y^2 > x$

(B) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, y^2 \leq x$

(C) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y^2 > x$

(D) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y^2 \leq x$

3. Let z_1 , z_2 and z_3 form an equilateral triangle as shown below.



Which of the following statements is true?

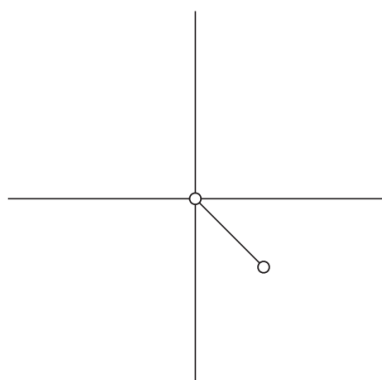
- (A) $\frac{z_2 - z_1}{z_1 - z_3} = \text{cis} \frac{\pi}{6}$
- (B) $\frac{z_2 - z_1}{z_3 - z_1} = \text{cis} \frac{\pi}{6}$
- (C) $\frac{z_2 - z_1}{z_1 - z_3} = \text{cis} \frac{\pi}{3}$
- (D) $\frac{z_2 - z_1}{z_3 - z_1} = \text{cis} \frac{\pi}{3}$
4. $z = x + iy$ is a complex number, such that it is NOT purely imaginary.

The equation of the locus of z such that $\text{Re}\left(z - \frac{1}{z}\right) = 0$ is:

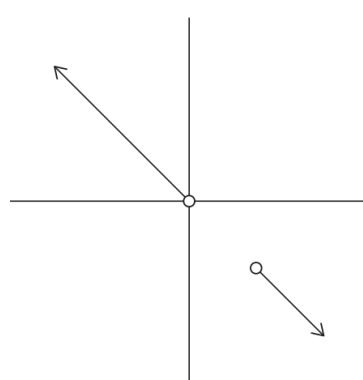
- (A) $x = 0$
- (B) $y = 0$
- (C) $x^2 + y^2 = 1$
- (D) $x + y - \frac{1}{x + y} = 0$

5. Which diagram best represents the solutions to the equation $\arg(z) = \arg(z + 1 - i)$?

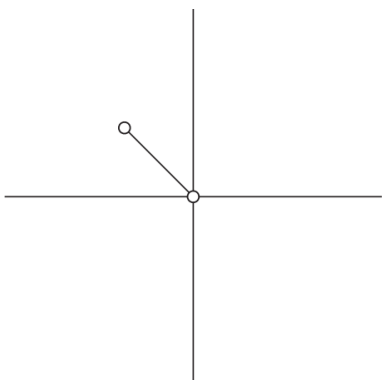
(A)



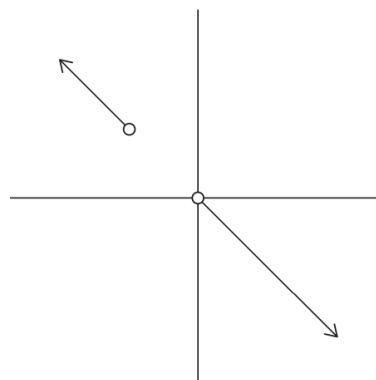
(B)



(C)



(D)



End of Section I

Section II

35 marks

Attempt Questions 6–9

Allow about 52 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (9 marks) Use a SEPARATE writing booklet.

- (a) Given two complex numbers, $z = 2 + 5i$ and $w = -1 + 2i$, find:
- | | | |
|-------|---------------------|---|
| (i) | $z - 2w$ | 1 |
| (ii) | zw | 1 |
| (iii) | $\overline{z + iw}$ | 2 |
- (b) It is given that $P(z) = z^3 - 4z^2 + 14z - 20$ has a root at $z = 1 + 3i$.
Factorise $P(z)$ fully over $\mathbb{R}$. 3
- (c) Prove by contradiction that $\log_3 5$ is irrational. 2

Question 7 (8 marks) Use a SEPARATE writing booklet.

- (a) Solve the equation $z^2 - (1 - 4i)z - 5 + i = 0$, leaving your answers in the form $a + ib$ where a and b are real. 3
- (b) (i) Sketch the following on the same Argand diagram: 3
- $$|z - 1 - 2i| = 2 \quad \text{and} \quad \arg(z - 1 - 2i) = \frac{\pi}{3}$$
- (ii) Find the complex number which represents the intersection of these two equations. 2

Question 8 (9 marks) Use a SEPARATE writing booklet.

- (a) Prove by contrapositive that if $3n + 7$ is even, then n is odd. 2
- (b) Let $z = 3\text{cis}\frac{\pi}{6}$ and $w = 2\text{cis}\frac{\pi}{4}$.
- (i) Express z and w in Cartesian form. 2
- (ii) Hence, express $\frac{w}{z}$ in both modulus-argument and Cartesian form. 3
- (iii) Hence, find the exact values of $\cos\frac{\pi}{12}$ and $\sin\frac{\pi}{12}$. 2

Question 9 (9 marks) Use a SEPARATE writing booklet.

- (a) On an Argand diagram, the points A, B, C and D represent the complex numbers α, β, γ , and δ respectively.
If $\alpha + \gamma = \beta + \delta$, deduce that $ABCD$ is a parallelogram. 2
- (b) Consider the following statement:
If n divides $(a + b)$ and n divides $(b + c)$, then n divides $(a + c)$.
Prove or disprove this statement. 2
- (c) (i) On an Argand diagram, sketch the curve defined by $\arg\left(\frac{z+2}{z-2}\right) = \frac{\pi}{3}$ 2
- (ii) Determine the Cartesian equation of this locus. 3

End of Paper

Section 1

1. $-1+i\sqrt{3} = z$

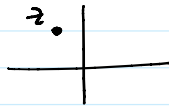
$$|z| = \sqrt{(-1)^2 + (\sqrt{3})^2}$$

$$= \sqrt{4}$$

$$= 2$$

$$\arg(z) = \tan^{-1}(\sqrt{3}/-1)$$

$$= \frac{2\pi}{3}$$



$$\therefore -1+i\sqrt{3} = 2 \operatorname{cis} 2\pi/3$$

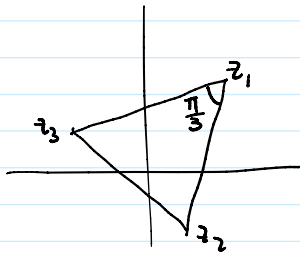
(C) ✓

2. P: $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y^2 > x$

$\neg P: \exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, y^2 \leq x$

(B) ✓

3.



(D) ✓

4. $z = x+iy \quad (x \neq 0)$

$$z - \frac{1}{z} = x+iy - \frac{1}{x+iy} \times \frac{x-iy}{x-iy}$$

$$= x+iy - \frac{x-iy}{x^2+y^2}$$

$$= x - \frac{x}{x^2+y^2} + i \left(y + \frac{y}{x^2+y^2} \right)$$

$$\operatorname{Re}\left(z - \frac{1}{z}\right) = 0$$

$$x - \frac{x}{x^2+y^2} = 0$$

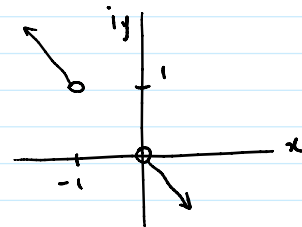
$$x = \frac{x}{x^2+y^2}$$

$$1 = \frac{1}{x^2+y^2} \quad (x \neq 0)$$

$$\therefore x^2+y^2 = 1$$

(C) ✓

5. $\arg(z) = \arg(z+1-i)$
 $= \arg(z - (-1+i))$



(D) ✓

Section 2

Question 6

(a) $z = 2 + 5i$, $w = -1 + 2i$

$$\begin{aligned} \text{(i)} \quad z - 2w &= 2 + 5i - 2(-1 + 2i) \\ &= 2 + 5i + 2 - 4i \\ &= 4 + i \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad zw &= (2 + 5i)(-1 + 2i) \\ &= -2 + 4i - 5i - 10 \\ &= -12 - i \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \overline{z + iw} &= \overline{2 + 5i + i(-1 + 2i)} \\ &= \overline{2 + 5i - i - 2} \\ &= \overline{4i} \\ &= -4i \end{aligned}$$

(b) $P(z) = z^3 - 4z^2 + 14z - 20$, root $z = 1 + 3i$

Since $P(z)$ has real coefficients, $\bar{z} = 1 - 3i$ is also a root ✓

Then, sum of roots $\alpha + \beta + \gamma = \frac{-b}{a}$

$$1 + 3i + 1 - 3i + \gamma = -\frac{4}{1}$$

$$2 + \gamma = 4$$

$$\gamma = 2$$

$$\begin{aligned} \therefore P(z) &= (z - 2)(z - (1 + 3i))(z - (1 - 3i)) \\ &= (z - 2)(z^2 - 2z + 10) \end{aligned}$$

(c) RTP: $\log_3 5$ is irrational

Suppose $\log_3 5$ is rational

i.e. $\log_3 5 = \frac{p}{q}$ where p, q are integers with no common factors except 1. ✓

$$q \log_3 5 = p$$

$$\log_3 5^q = p$$

$$5^q = 3^p, \text{ which is a contradiction since 3 and 5 are co-prime } \checkmark$$

$\therefore$ By contradiction, $\log_3 5$ is irrational.

Question 7

(a) $z^2 - (-1+4i)z - 5+i = 0$

$$\Delta = B^2 - 4AC$$

$$= (-1+4i)^2 - 4(1)(-5+i)$$

$$= 1 - 8i - 16 + 20 - 4i$$

$$= 5 - 12i$$

$$z = \frac{-(-1+4i) \pm \sqrt{5-12i}}{2(1)}$$

$$= \frac{1-4i \pm \sqrt{5-12i}}{2}$$

$$= \frac{1-4i \pm (3-2i)}{2}$$

$$= \frac{1-4i+3-2i}{2}, \frac{1-4i-3+2i}{2}$$

$$= \frac{4-6i}{2}, \frac{-2-2i}{2}$$

$$= 2-3i, -1-i$$

Let $(x+iy)^2 = 5-12i$

$$x^2 + 2ixy - y^2 = 5-12i$$

$$\begin{cases} x^2 - y^2 = 5 & \text{--- (1)} \\ 2xy = -12 & \text{--- (2)} \end{cases}$$

$$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$$

$$= 5^2 + (-12)^2$$

$$= 169$$

$$x^2 + y^2 = 13 \quad \text{--- (3)}$$

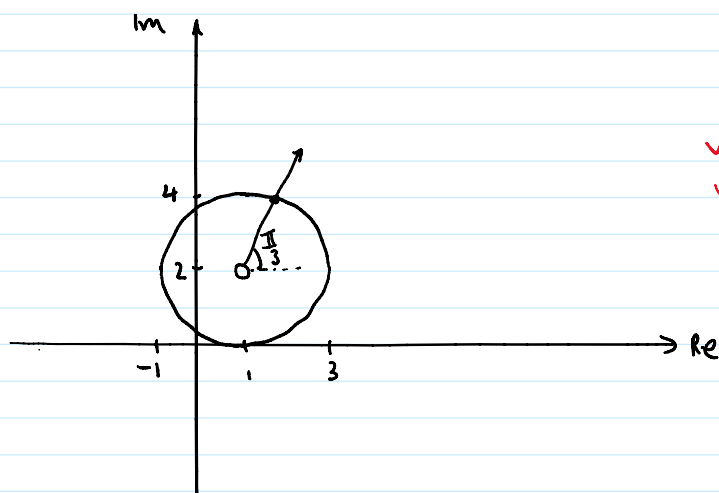
$$\text{(1) + (3)}$$

$$2x^2 = 18$$

$$x^2 = 9$$

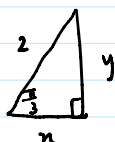
$$x = \pm 3 \rightarrow \text{Sub in (2)} \quad y = \mp 2$$

(b) (i) $|z-1-2i| = 2$ and $\arg(z-1-2i) = \frac{\pi}{3}$
 $|z-(1+2i)| = 2$ $\arg(z-(1+2i)) = \frac{\pi}{3}$



✓ circle centre
 ✓ circle radius
 ✓ Ray

(ii)



$$\cos \frac{\pi}{3} = \frac{x}{2}$$

$$\frac{1}{2} = \frac{x}{2}$$

$$x = 1, y = \sqrt{3}$$

$$\therefore \text{POI at } z = 2+i(2+\sqrt{3})$$

Question 8

(a) RTP: If $3n+7$ is even, then n is odd

Suppose n is even, i.e. $n=2k$, $k \in \mathbb{Z}$ ✓

$$\begin{aligned}\text{Then } 3n+7 &= 3(2k)+7 \\ &= 6k+7 \\ &= 2(3k+3)+1 \\ &= 2A+1, \text{ which is odd } (A \in \mathbb{Z})\end{aligned}$$
 ✓

$\therefore$ If n is even, then $3n+7$ is odd

$\therefore$ By contrapositive, if $3n+7$ is even, then n is odd

(b) $z = 3\text{cis}\frac{\pi}{6}$, $w = 2\text{cis}\frac{\pi}{4}$

$$\begin{aligned}\text{(i) } z &= 3(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}) \\ &= 3(\sqrt{3}/2 + i \cdot 1/2) \\ &= \frac{3\sqrt{3}}{2} + i \frac{3}{2}\end{aligned}$$
 ✓

$$\begin{aligned}w &= 2(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}) \\ &= 2(\frac{1}{\sqrt{2}} + i \cdot \frac{1}{\sqrt{2}}) \\ &= \frac{2}{\sqrt{2}} + i \frac{2}{\sqrt{2}} \\ &= \sqrt{2} + i\sqrt{2}\end{aligned}$$
 ✓

$$\begin{aligned}\text{(ii) } \frac{w}{z} &= \frac{2\text{cis}\frac{\pi}{4}}{3\text{cis}\frac{\pi}{6}} \\ &= \frac{2}{3} \text{cis}\left(\frac{\pi}{4} - \frac{\pi}{6}\right) \\ &= \frac{2}{3} \text{cis}\frac{\pi}{12}\end{aligned}$$
 ✓

$$\begin{aligned}\frac{w}{z} &= \frac{\sqrt{2} + i\sqrt{2}}{\frac{3\sqrt{3}}{2} + i \frac{3}{2}} \\ &= \frac{2}{3} \cdot \frac{\sqrt{2} + i\sqrt{2}}{\sqrt{3} + i} \times \frac{\sqrt{3} - i}{\sqrt{3} - i} \\ &= \frac{2}{3} \cdot \frac{\sqrt{6} - i\sqrt{2} + i\sqrt{6} + \sqrt{2}}{3+1} \\ &= \frac{2}{3} \cdot \frac{(\sqrt{6} + \sqrt{2}) + i(\sqrt{6} - \sqrt{2})}{4} \\ &= \frac{1}{6} [\sqrt{6} + \sqrt{2} + i(\sqrt{6} - \sqrt{2})] \\ &= \frac{\sqrt{6} + \sqrt{2}}{6} + i \frac{\sqrt{6} - \sqrt{2}}{6}\end{aligned}$$
 ✓

(iii) Equating real & imaginary parts of $\frac{w}{z}$

$$\frac{2}{3} \cos \frac{\pi}{12} = \frac{\sqrt{6} + \sqrt{2}}{6}$$

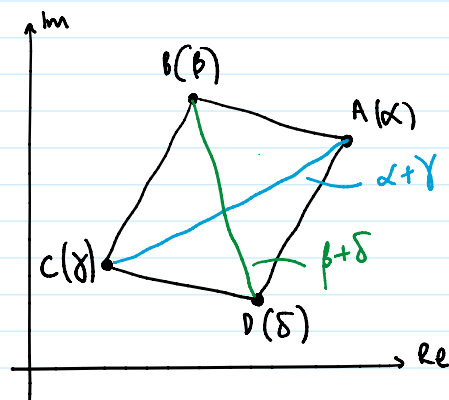
$$\therefore \cos \frac{\pi}{12} = \frac{\sqrt{6} + \sqrt{2}}{4}$$
 ✓

$$\frac{2}{3} \sin \frac{\pi}{12} = \frac{\sqrt{6} - \sqrt{2}}{6}$$

$$\therefore \sin \frac{\pi}{12} = \frac{\sqrt{6} - \sqrt{2}}{4}$$
 ✓

Question 9

(a) A, B, C, D represent $\alpha, \beta, \gamma, \delta$



$$\alpha + \gamma = \beta + \delta$$

$$\frac{1}{2}(\alpha + \gamma) = \frac{1}{2}(\beta + \delta)$$

$\therefore$ Midpoints of AC and BD are the same

$\therefore$ Diagonals AC and BD bisect each other

$\therefore$ ABCD is a parallelogram.

b) RTP(?): If n divides $(a+b)$ and n divides $(b+c)$ then n divides $(a+c)$.

Let $\underbrace{a+b}_{(1)} = nP$ and $\underbrace{b+c}_{(2)} = nQ$, $P, Q \in \mathbb{Z}$

① + ②

$$a + 2b + c = nQ + nP$$

$$a + c = n(P+Q) - 2b, \text{ which isn't divisible by } n?$$

} not necessary

try counter example to disprove

Let $a=4$, $b=8$, $c=7$, $n=3$

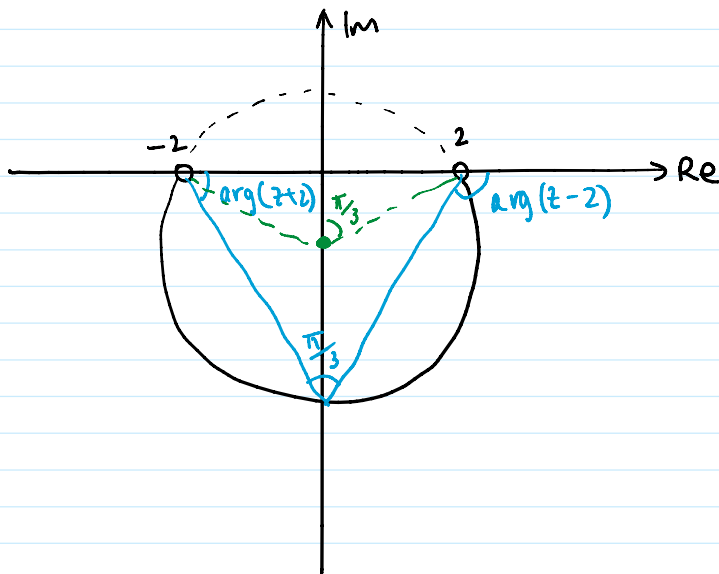
$$a+b=4+8=12, \text{ divisible by } 3$$

$$b+c=8+7=15, \text{ divisible by } 3$$

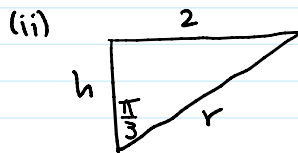
$$a+c=4+7=11, \text{ not divisible by } 3$$

$\therefore$ By counter example, the statement is false.

(c) (i) $\arg\left(\frac{z+2}{z-2}\right) = \frac{\pi}{3}$
 $\arg(z+2) - \arg(z-2) = \frac{\pi}{3}$



✓ major arc incl. open circles
 ✓ showing blue angles



$$\sin \frac{\pi}{3} = \frac{2}{r}$$

$$\frac{\sqrt{3}}{2} = \frac{2}{r}$$

$$r = \frac{4}{\sqrt{3}} \quad \checkmark$$

$$\tan \frac{\pi}{3} = \frac{2}{h}$$

$$\sqrt{3} = \frac{2}{h}$$

$$h = \frac{2}{\sqrt{3}} \quad \checkmark$$

Circle with centre $(0, -\frac{2}{3})$ and radius $\frac{4}{\sqrt{3}}$ units

$$\therefore x^2 + \left(y + \frac{2}{3}\right)^2 = \frac{16}{3} \quad \checkmark$$